



CMPT 413/713: Natural Language Processing

Language Models

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Adapted from slides from Anoop Sarkar, Danqi Chen and Karthik Narasimhan

Consider

Today, in Vancouver, it is 31 F and red

vs

Today, in Vancouver, it is 31 F and snowing

- Both are grammatical
- But which is more likely?

What is Language Modeling?

- We want to be able to estimate the **probability** of a **sequence of words**
- How likely is a given phrase / sentence / paragraph / document?
- We want to be able to compute $P(s) = P(w_1, w_2, \dots, w_T)$

$$P(s) = P(w_1, \dots, w_T) = \prod_{t=1}^T P(w_t | w_{<t})$$

Why is this useful?

Applications

- Predicting words is important in many situations
 - Autocomplete
 - Machine translation

$$P(\text{a smooth finish}) > P(\text{a flat finish})$$

- Speech recognition/Spell checking

$$P(\text{high school principal}) > P(\text{high school principle})$$

- Information extraction, Question answering



Hypothesis scores for speech recognition

From acoustic signal to candidate transcriptions

Hypothesis	Score
the station signs are in deep in english	-14732
the stations signs are in deep in english	-14735
the station signs are in deep into english	-14739
the station 's signs are in deep in english	-14740
the station signs are in deep in the english	-14741
the station signs are indeed in english	-14757
the station 's signs are indeed in english	-14760
the station signs are indians in english	-14790
the station signs are indian in english	-14799
the stations signs are indians in english	-14807
the stations signs are indians and english	-14815

Hypothesis scores for machine translation

From source language to target language candidates

Hypothesis	Score
we must also discuss a vision .	-29.63
we must also discuss on a vision .	-31.58
it is also discuss a vision .	-31.96
we must discuss on greater vision .	-36.09
⋮	⋮

Why is language model important?

- Much of the current successes in NLP comes from large pre-trained language models (BERT, GPT, T5, ...)
- By training neural language models, we can obtain useful representations for words and sentences.
- Can take the pre-trained language fine tune for specific tasks or use in zero-shot setting



matte painting of a bonsai tree;
trending on artstation.



Language Modeling

Predict probability of sequence of words

$$P(s) = P(w_1, \dots, w_T) = \prod_{t=1}^T P(w_t | w_{<t})$$

with n-grams

$$P(w_t | w_{<t}) \approx P(w_t | w_{t-n+1, t-1})$$

with HMMs

$$P(w_t | w_{<t}) \approx P(w_t | h_t) P(h_t | h_{t-n+1, t-1})$$

with neural networks

$$p(w_t | w_{<t}) \approx p(w_t | \phi(w_1, \dots, w_{t-1}))$$

with fixed window

$$P(w_t | w_{<t}) \approx P(w_t | \phi(w_{t-n+1, t-1}))$$

with RNNs

$$P(w_t | w_{<t}) \approx P(w_t | \mathbf{h}_t), \mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{x}_t)$$

What to know about LMs?

- What is a language model?
- Statistical language model using **ngrams**
- How to build a language model? **Training the model from data**
Learning/estimating model parameters
- MLE and smoothing
- How to use the language model? **Generation**
- How to tell if our language model is working well? **Evaluation**

What is a language model?

Probabilistic model of a sequence of words

Setup: Assume a finite vocabulary of words V

$V = \{\text{cat, clown, crazy, killer, mat, on, sat, the}\}$

V can be used to construct an infinite set of sentences (sequences of words)

$V^+ = \{\text{clown, cat sat, killer clown, crazy clown, crazy cat, crazy killer clown, killer crazy clown, ...}\}$

where a sentence is defined as $s \in V^+$ where $s = \{w_1, \dots, w_n\}$



What is a language model?

Probabilistic model of a sequence of words

Given a **training data** set of example sentences

$$S = \{s_1, s_2, \dots, s_m\}, s_i \in V^+$$

Estimate a probability model

$$\sum_{s_j \in V^+} P(s_j) = \sum_j P(w_1, \dots, w_{n_j}) = 1.0$$

Language Model

- $p(\text{clown}) = 1\text{e-}5$
- $p(\text{killer}) = 1\text{e-}6$
- $p(\text{killer clown}) = 1\text{e-}12$
- $p(\text{crazy killer clown}) = 1\text{e-}21$
- $p(\text{crazy killer clown killer}) = 1\text{e-}110$
- $p(\text{crazy slow killer killer}) = 1\text{e-}127$
- ...

Where do we get the vocabulary?

Common Setup: Assume a finite vocabulary of words V

- Get from a list of words (say a dictionary)
- Build from training data
- Decide on vocabulary size (say $|V| = 50K$) and then pick most frequent words
- Take words that occur more than T times.



Learning language models

How to estimate the probability of a sentence?

- We can **directly count** using a training data set of sentences

$$P(w_1, \dots, w_n) = \frac{C(w_1, \dots, w_n)}{N}$$

Problem: does not generalize to new sentences **unseen** in the training data

- C is a function that counts how many times each sentence occurs
- N is the sum over all possible $C(\cdot)$ values

Estimating joint probabilities with the chain rule

$$\begin{aligned} P(w_1, w_2, \dots, w_n) &= P(w_1)P(w_2|w_1)P(w_3|w_1, w_2) \times \dots \times P(w_n|w_1, w_2, \dots, w_{n-1}) \\ &= \prod_{i=1}^n P(w_i|w_1, \dots, w_{i-1}) \end{aligned}$$

Example

Sentence: “the cat sat on the mat”

$$\begin{aligned} P(\text{the cat sat on the mat}) &= P(\text{the}) \times P(\text{cat}|\text{the}) \times P(\text{sat}|\text{the cat}) \\ &\quad \times P(\text{on}|\text{the cat sat}) \times P(\text{the}|\text{the cat sat on}) \\ &\quad \times P(\text{mat}|\text{the cat sat on the}) \end{aligned}$$

Markov assumption

- Use only the **recent past** to predict the next word
- Reduces the number of estimated parameters in exchange for modeling capacity

- 0th order

$$P(\text{mat}|\text{the cat sat on the}) \approx P(\text{mat})$$

- 1st order

$$P(\text{mat}|\text{the cat sat on the}) \approx P(\text{mat}|\text{the})$$

- 2nd order $P(w_i | w_1 w_2 \dots w_{i-1}) \approx P(w_i | w_{i-k} \dots w_{i-1})$

Markov assumption

- Use only the recent past to predict the next word
- Reduces the number of estimated parameters in exchange for modeling capacity
- 0th order $P(\text{mat}|\text{the cat sat on the}) \approx P(\text{mat})$
- 1st order $P(\text{mat}|\text{the cat sat on the}) \approx P(\text{mat}|\text{the})$
- 2nd order $P(\text{mat}|\text{the cat sat on the}) \approx P(\text{mat}|\text{on the})$
- kth order $P(w_i | w_1 w_2 \dots w_{i-1}) \approx P(w_i | w_{i-k} \dots w_{i-1})$
- Probability of sequence: $P(w_1 w_2 \dots w_n) \approx \prod_i P(w_i | w_{i-k} \dots w_{i-1})$

Estimating n-gram probabilities

- **Maximum likelihood estimate (MLE):** Use counts from text corpus

Unigram
$$P(w_i) = \frac{C(w_i)}{\sum_{w_i \in V} C(w_i)} = \frac{C(w_i)}{N}$$

Bigram
$$P(w_i | w_{i-1}) = \frac{C(w_{i-1}, w_i)}{C(w_{i-1})}$$

Trigram
$$P(w_i | w_{i-1}, w_{i-2}) = \frac{C(w_{i-2}, w_{i-1}, w_i)}{C(w_{i-2}, w_{i-1})}$$

Can reuse counts for multiple estimations

Maximum Likelihood Estimation

We want to find the set of parameters $\hat{\theta}$ that maximize the probability of the training data

$$\hat{\theta} = \arg \max_{\theta \in \Theta} \hat{L}(\theta; \mathcal{D})$$

Parameters

$$\theta : \{p(w_i | w_1, \dots, w_{i-1})\}$$

Corpus of N sentences

Using our model, we can estimate the probabilities of these sentences

Likelihood

$$\hat{L}_m(\theta; \mathcal{D}) = \prod_{j=1}^m P(w_1^{(j)}, \dots, w_{n_j}^{(j)}) = \prod_{j=1}^m \prod_{i=1}^{n_j} P(w_i^{(j)} | w_1^{(j)}, \dots, w_{n_j}^{(j)})$$

Log-Likelihood

$$\hat{\ell}_m(\theta; \mathcal{D}) = \sum_{j=1}^m \log P(w_1^{(j)}, \dots, w_{n_j}^{(j)}) = \sum_{j=1}^m \sum_{i=1}^{n_j} \log P(w_i^{(j)} | w_1^{(j)}, \dots, w_{n_j}^{(j)})$$

Easier to work with (products to sums)
Numeric underflow less of a issue

Likelihood function

- How likely it is to see the examples in the training data
- Function of the parameters you are using to model the probability
- Probability density over data samples (sentences)

Typical assumptions

- Samples are iid (independently and identically distributed)

Maximum Likelihood Estimation (for categorical distributions)

- Unigram: $P(w)$
- Probability: $P(w) \geq 0, \sum_{w \in V} P(w) = 1$
- MLE is the **sample mean**
- Optimize $P_{\text{ML}}(w) = \arg \max_{P(w)} \sum_{j=1}^N \sum_{i=1}^{n_j} \log P(w_i^{(j)}) = \arg \max_{P(w)} \sum_{w \in V} C(w) \log P(w)$
- Solve using Lagrange Multipliers $g(\lambda, P(\cdot)) = \sum_{w \in V} C(w) \log P(w) - \lambda (\sum_{w \in V} P(w) - 1)$

$$P(w) = \frac{C(w)}{\lambda} \quad \longrightarrow \quad P(w) = \frac{C(w)}{\sum_{w \in V} C(w)} = \frac{C(w)}{N}$$

Using Language Models

How to use n-gram LMs?

- Computing probability

$$P(\text{high school } \mathbf{principal}) > P(\text{high school } \mathbf{principle})$$

- Completion

$$\arg \max_{w \in V} P(w | \text{where, is, SFU})$$

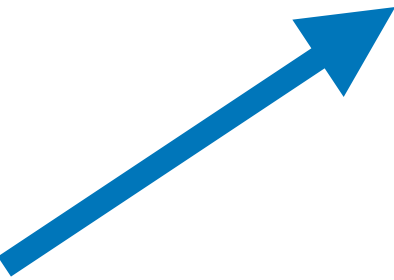
- Generating text

Computing the probability of a sentence

Apply the Chain Rule: the trigram model

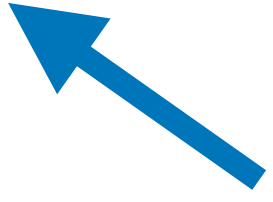
$$\begin{aligned} P(w_1, \dots, w_n) \\ &\approx P(w_1)P(w_2 \mid w_1)P(w_3 \mid w_1, w_2) \dots P(w_n \mid w_{n-2}, w_{n-1}) \\ &\approx P(w_1)P(w_2 \mid w_1) \prod_{i=3}^n P(w_i \mid w_{i-2}, w_{i-1}) \end{aligned}$$

Not trigrams!
Pad our sentence
with <s>
(start of sentence
markers)



$$P(w_1, \dots, w_n) \approx \prod_{i=1}^n P(w_i \mid w_{i-2}, w_{i-1})$$

Not proper
distribution unless we
add a **stop** symbol
(or </s> end of
sentence marker)



$$P(w_1) = P(w_1 \mid \langle s \rangle \langle s \rangle)$$

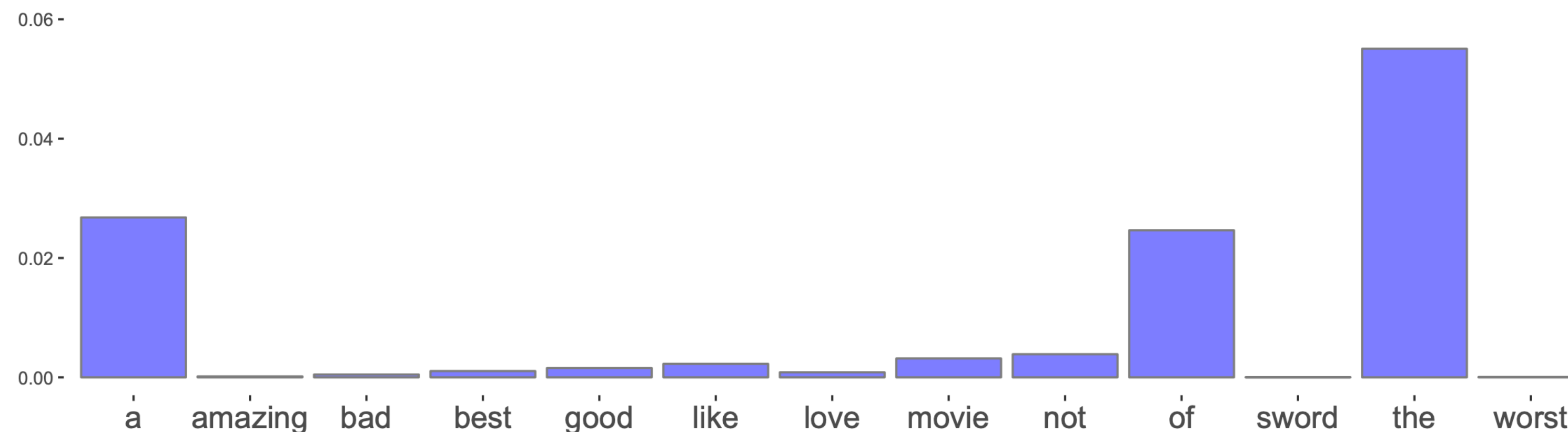
$$P(w_2 \mid w_1) = P(w_2 \mid \langle s \rangle w_1)$$

Generating text from n-grams

generative model

- How do you generate text from an n-gram model?
- Sample from distribution, generating tokens from left to right
- Use the last $n - 1$ words for context
- Select from multinomial over the vocabulary that include a **STOP** token. Repeat until **STOP** is generated.

context1	context2	generated word
START	START	The
START	The	dog
The	dog	walked
dog	walked	in



Generalization

Number of Parameters

How many probabilities in each n -gram model

$$\mathcal{V} = \{\text{cat}, \text{crazy}, \text{mat}, \text{sat}\}$$

Question

How many unigram probabilities: $P(x)$ for $x \in \mathcal{V}$?

Number of Parameters

How many probabilities in each n -gram model

$$\mathcal{V} = \{\text{cat}, \text{crazy}, \text{mat}, \text{sat}\}$$

Question

How many bigram probabilities: $P(y|x)$ for $x, y \in \mathcal{V}$?

Number of Parameters

How many probabilities in each n -gram model

$$\mathcal{V} = \{\text{cat}, \text{crazy}, \text{mat}, \text{sat}\}$$

Question

How many trigram probabilities: $P(z|x, y)$ for $x, y, z \in \mathcal{V}$?

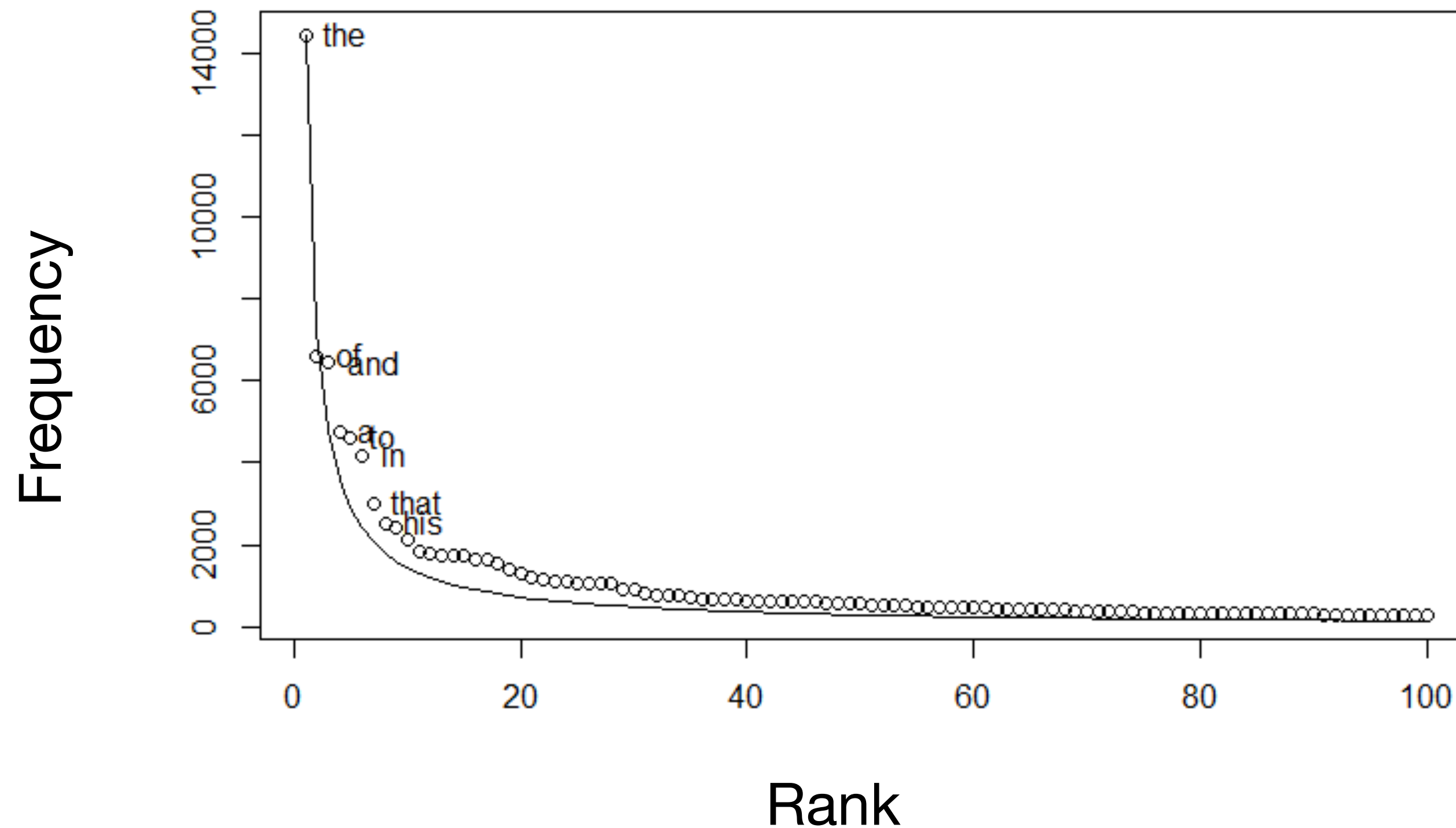
Number of parameters

- ▶ Assume $|\mathcal{V}| = 50,000$ (a realistic vocabulary size for English)
- ▶ What is the minimum size of training data in tokens?
 - ▶ If you wanted to observe all unigrams at least once.
 - ▶ If you wanted to observe all trigrams at least once.

Generalization of n-grams

- Not all n-grams will be observed in training data!
- There can be unknown words in the test set!
- Test corpus might have some that have zero probability under our model
- Training set: *Google news*
- Test set: *Shakespeare*
- $P(\text{affray} \mid \text{voice doth us}) = 0 \quad \Rightarrow \quad P(\text{test corpus}) = 0$

Sparsity in language



$$freq \propto \frac{1}{rank}$$

Zipf's Law

- Long tail of infrequent words
- Most finite-size corpora will have this problem.

Unknown words

- Typically assume closed vocabulary
- What about words not in the vocabulary?
 - Known as OOV (out-of-vocabulary) words.
 - Introduce **<UNK> token** to represent the unknown words
- If we never see these words in the training data, any sentence with these words will get a probability of 0!
- Can handle these unknown words by:
 - Estimate the probability of unknown word as: $P_{\text{unk}}(w) = \frac{1}{|V_{\text{all}}|}$ $V_{\text{all}} = V \cup \{< \text{UNK} >\}$
 - Or modify training data so rare words (words that appear $< T$ times) are treated as **<UNK>**

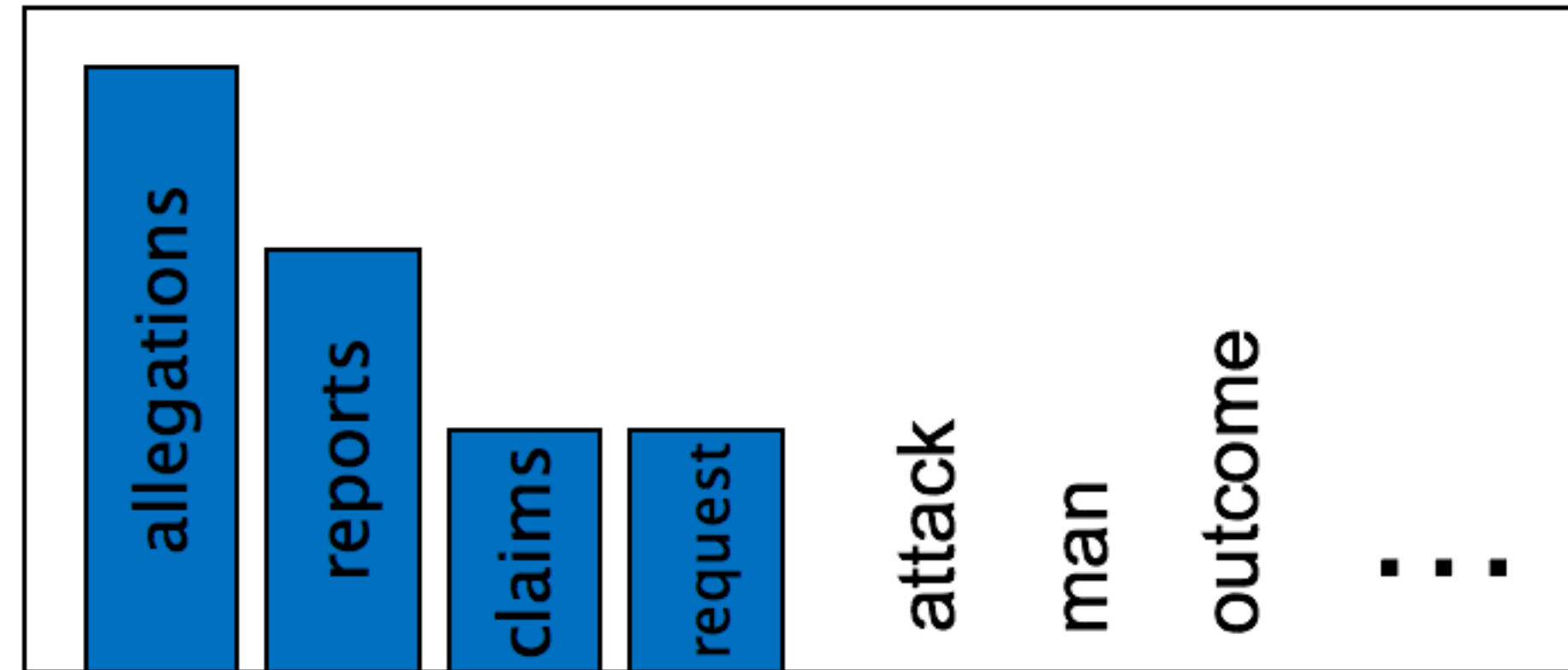
Smoothing n-gram Models

Smoothing intuition

Taking from the rich and giving to the poor

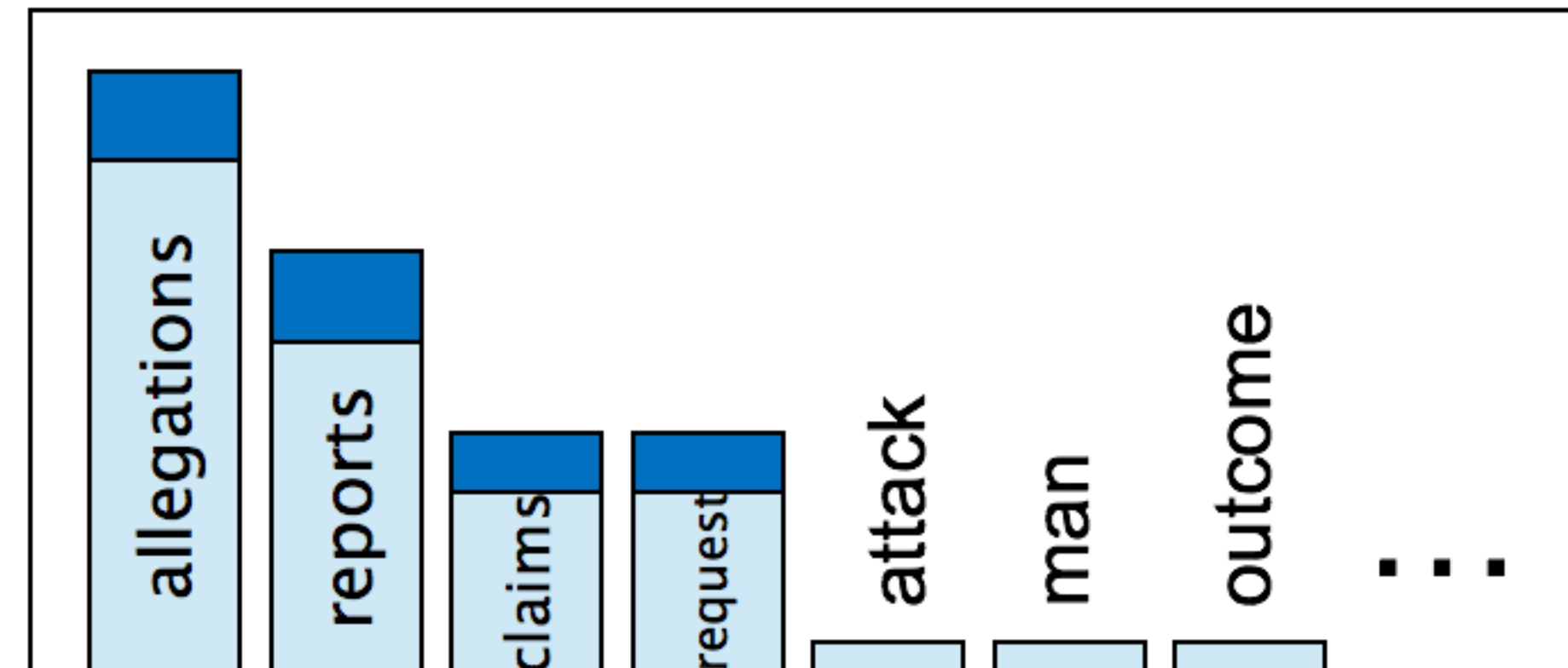
When we have sparse statistics:

$P(w \mid \text{denied the})$
3 allegations
2 reports
1 claims
1 request
7 total



Steal probability mass to generalize better

$P(w \mid \text{denied the})$
2.5 allegations
1.5 reports
0.5 claims
0.5 request
2 other
7 total



(Credits: Dan Klein)

Smoothing

- **Smoothing** deals with events that have been observed zero or very few times
- Handle sparsity by making sure all **probabilities are non-zero** in our model
- **Additive**: Add a small amount to all probabilities
- **Discounting**: Redistribute probability mass from observed n-grams to unobserved ones
- **Interpolation**: Use a combination of different n-grams
- **Back-off**: Use lower order n-grams if higher ones are too sparse

Ensure proper probability distribution

Add-one (Laplace) smoothing

- Why add 1? 1 is an overestimate for unobserved events

- Max likelihood estimate for bigrams: $P_{\text{ML}}(w_i | w_{i-1}) = \frac{C(w_{i-1}, w_i)}{C(w_{i-1})}$

- Let $|V|$ be the number of words in our vocabulary. Assign count of 1 to unseen bigrams

- After smoothing:

$$P_{\text{Add1}}(w_i | w_{i-1}) = \frac{1 + C(w_{i-1}, w_i)}{|V| + C(w_{i-1})}$$

Additive smoothing

(Lidstone 1920, Jeffreys 1948)

- Why add 1? 1 is an overestimate for unobserved events

$$P_{\text{Add1}}(w_i | w_{i-1}) = \frac{1 + C(w_{i-1}, w_i)}{|V| + C(w_{i-1})}$$

- Additive smoothing ($0 < \delta \leq 1$):

$$P_{\text{Add}\delta}(w_i | w_{i-1}) = \frac{\delta + C(w_{i-1}, w_i)}{\delta \times |V| + C(w_{i-1})}$$

- Also known as add-alpha (the symbol α is used instead of δ)

Raw bigram counts (Berkeley restaurant corpus)

$$P_{\text{ML}}(w_i | w_{i-1}) = \frac{C(w_{i-1}, w_i)}{C(w_{i-1})}$$

- Out of 9222 sentences

	i	want	to	eat	chinese	food	lunch	spend
i	5	827	0	9	0	0	0	2
want	2	0	608	1	6	6	5	1
to	2	0	4	686	2	0	6	211
eat	0	0	2	0	16	2	42	0
chinese	1	0	0	0	0	82	1	0
food	15	0	15	0	1	4	0	0
lunch	2	0	0	0	0	1	0	0
spend	1	0	1	0	0	0	0	0

(Credits: Dan Jurafsky)

Smoothed bigram counts

	i	want	to	eat	chinese	food	lunch	spend
i	6	828	1	10	1	1	1	3
want	3	1	609	2	7	7	6	2
to	3	1	5	687	3	1	7	212
eat	1	1	3	1	17	3	43	1
chinese	2	1	1	1	1	83	2	1
food	16	1	16	1	2	5	1	1
lunch	3	1	1	1	1	2	1	1
spend	2	1	2	1	1	1	1	1

Smoothed bigram probabilities

(Laplace Add-1 smoothing)

$$P_{\text{Add1}}(w_i | w_{i-1}) = \frac{1 + C(w_{i-1}, w_i)}{|V| + C(w_{i-1})}$$

	i	want	to	eat	chinese	food	lunch	spend
i	0.0015	0.21	0.00025	0.0025	0.00025	0.00025	0.00025	0.00075
want	0.0013	0.00042	0.26	0.00084	0.0029	0.0029	0.0025	0.00084
to	0.00078	0.00026	0.0013	0.18	0.00078	0.00026	0.0018	0.055
eat	0.00046	0.00046	0.0014	0.00046	0.0078	0.0014	0.02	0.00046
chinese	0.0012	0.00062	0.00062	0.00062	0.00062	0.052	0.0012	0.00062
food	0.0063	0.00039	0.0063	0.00039	0.00079	0.002	0.00039	0.00039
lunch	0.0017	0.00056	0.00056	0.00056	0.00056	0.0011	0.00056	0.00056
spend	0.0012	0.00058	0.0012	0.00058	0.00058	0.00058	0.00058	0.00058

(Credits: Dan Jurafsky)

The problem with Laplace smoothing

Too much discounted
from popular words!

Raw
counts

	i	want	to	eat	chinese	food	lunch	spend
i	5	827	0	9	0	0	0	2
want	2	0	608	1	6	6	5	1
to	2	0	4	686	2	0	6	211
eat	0	0	2	0	16	2	42	0
chinese	1	0	0	0	0	82	1	0
food	15	0	15	0	1	4	0	0
lunch	2	0	0	0	0	1	0	0
spend	1	0	1	0	0	0	0	0

Reconstituted
counts

	i	want	to	eat	chinese	food	lunch	spend
i	3.8	527	0.64	6.4	0.64	0.64	0.64	1.9
want	1.2	0.39	238	0.78	2.7	2.7	2.3	0.78
to	1.9	0.63	3.1	430	1.9	0.63	4.4	133
eat	0.34	0.34	1	0.34	5.8	1	15	0.34
chinese	0.2	0.098	0.098	0.098	0.098	8.2	0.2	0.098
food	6.9	0.43	6.9	0.43	0.86	2.2	0.43	0.43
lunch	0.57	0.19	0.19	0.19	0.19	0.38	0.19	0.19
spend	0.32	0.16	0.32	0.16	0.16	0.16	0.16	0.16

$$c^*(w_{n-1}w_n) = \frac{[C(w_{n-1}w_n) + 1] \times C(w_{n-1})}{C(w_{n-1}) + V}$$

(Credits: Dan Jurafsky)

Linear Interpolation (Jelinek-Mercer Smoothing)

$$\begin{aligned} P_{\text{interp}}(w_i | w_{i-1}, w_{i-2}) = & \lambda_1 P(w_i | w_{i-1}, w_{i-2}) \\ & + \lambda_2 P(w_i | w_{i-1}) \\ & + \lambda_3 P(w_i) \\ & + \lambda_4 \frac{1}{|V|} \end{aligned}$$
$$\sum_i \lambda_i = 1$$

- Use a combination of models to estimate probability
- Strong empirical performance

Jelinek and Mercer (1980)

Linear Interpolation (Jelinek-Mercer Smoothing)

- It's also possible to formulate the interpolation in a recursive manner:

$$P_{\text{JM}}(n\text{gram}) = \lambda_n P_{\text{ML}}(n\text{gram}) + (1 - \lambda_n) P_{\text{JM}}(n - 1\text{gram})$$

$$P_{\text{JM}}(w_i | w_{i-n+1}^{i-1}) = \lambda_n P_{\text{ML}}(w_i | w_{i-n+1}^{i-1}) + (1 - \lambda_n) P_{\text{JM}}(w_i | w_{i-n+2}^{i-1})$$

$$P_{\text{JM}}(w_i) = \lambda_1 P_{\text{ML}}(w_i) + (1 - \lambda_1) \frac{\delta}{|V|}$$

Brown et al (1992)

Linear Interpolation: Finding lambda

$$P_{\text{JM}}(\text{ngram}) = \lambda P_{\text{ML}}(\text{ngram}) + (1 - \lambda) P_{\text{JM}}(n - 1\text{gram})$$

- Interpolation parameters (λ) are hyper parameters. Tune them on the “held-out” set.



- Improved JM smoothing, a different λ for each w_i

$$P_{\text{JM}}(w_i | w_{i-1}) = \lambda(w_{i-1}) P_{\text{ML}}(w_i | w_{i-1}) + (1 - \lambda(w_{i-1})) P_{\text{JM}}(w_i)$$

Discounting

Bigram count in training	Bigram count in heldout set
0	.0000270
1	0.448
2	1.25
3	2.24
4	3.23
5	4.21
6	5.23
7	6.21
8	7.21
9	8.26

- Determine some “mass” to remove from probability estimates
- Redistribute mass among unseen n-grams
- Just choose an absolute value (D) to discount

more properly

$$\max(c(w_{i-1}, w_i) - D, 0)$$

α is set so the resulting probability values sums to one

Interpolated
absolute
discounting

$$P_{\text{absdis-i}}(w_i | w_{i-1}) = \frac{C(w_{i-1}, w_i) - D}{C(w_{i-1})} + \alpha(w_{i-1}) P_{\text{absdis-i}}(w_i)$$

Very similar to
Interpolated
Knesser-Ney

With interpolation, can also be with “backoff” as we will see

Interpolated Knesser-Ney

Popular state of the art n-gram smoothing

Cleverer count (based on number of contexts)

Modified Knesser-Ney: different discounting values

$$P_{\text{KN-i}}(w_i | w_{i-1}) = \frac{\max(C_{\text{KN}}(w_{i-1}, w_i) - D)}{\sum_{w'} C_{\text{KN}}(w_{i-1}, w')} + \alpha(w_{i-1}) P_{\text{KN-i}}(w_i)$$

more properly

$\max(c(w_{i-1}, w_i) - D, 0)$

α is set so the resulting probability values sums to one

Interpolated absolute discounting

$$P_{\text{absdis-i}}(w_i | w_{i-1}) = \frac{C(w_{i-1}, w_i) - D}{C(w_{i-1})} + \alpha(w_{i-1}) P_{\text{absdis-i}}(w_i)$$

Very similar to Interpolated Knesser-Ney

With interpolation, can also be with “backoff” as we will see

Back-off

- Use **n-gram** if enough evidence, else back off to **(n-1)-gram**

$$P_{\text{bo}}(w_i | w_{i-n+1}^{i-1}) \\ = \begin{cases} d_{w_{i-n+1}^i} \frac{C(w_{i-n+1}^i)}{C(w_{i-n+1}^{i-1})} & \text{if } C(w_{i-n+1}^i) > 0 \\ \alpha_{w_{i-n+1}^{i-1}} P_{\text{bo}}(w_i | w_{i-n+2}^{i-1}) & \text{otherwise} \end{cases}$$

- d = amount of discounting (Katz back-off, 1987)
- α = back-off weight

Backoff Smoothing with Discounting

- Absolute Discounting with backoff (Ney, Essen, Knessler)

$$P_{\text{absdis-bo}}(w_i | w_{i-1}) = \begin{cases} \frac{C(w_{i-1} w_i) - D}{C(w_{i-1})} & \text{if } C(w_{i-1} w_i) > 0 \\ \alpha(w_{i-1}) P_{\text{absdis-bo}}(w_i) & \text{otherwise} \end{cases}$$

- Where $\alpha(w_{i-1})$ is chosen to ensure that $P_{\text{abs}}(w_i | w_{i-1})$ is a proper probability

Similar to
Backoff
Knessler-Ney,
1994

$$\alpha(w_{i-1}) = 1 - \sum_{w_i} \frac{C(w_{i-1} w_i) - D}{C(w_i)}$$

Different value of α for
each context word w_{i-1}

Backoff Smoothing with Discounting

- Let $D = 0.5$

- Missing probability mass:

$$\alpha(w_{i-1}) = 1 - \sum_{w_i} \frac{C(w_{i-1}w_i) - D}{C(w_i)}$$

$$\alpha(\text{the}) = 10 \times 0.5/48 = 5/48$$

- Divide this mass between words w for which the counts: $C(\text{the}, w) = 0$

x	$c(x)$	$c(x) - D$	$\frac{c(x) - D}{c(\text{the})}$
the	48		
the,dog	15	14.5	14.5/48
the,woman	11	10.5	10.5/48
the,man	10	9.5	9.5/48
the,park	5	4.5	4.5/48
the,job	2	1.5	1.5/48
the,telescope	1	0.5	0.5/48
the,manual	1	0.5	0.5/48
the,afternoon	1	0.5	0.5/48
the,country	1	0.5	0.5/48
the,street	1	0.5	0.5/48
TOTAL			0.8958
the,UNK	0		0.1042

Web-scale N-grams Smoothing

Keeping track of everything gets complicated

Not even a proper distribution!

- “Stupid backoff” (Brants et al, 2007)

$$S(w_i | w_{i-n+1}^{i-1}) = \begin{cases} \frac{C(w_{i-n+1}^i)}{C(w_{i-n+1}^{i-1})} & \text{if } C(w_{i-n+1}^i) > 0 \\ 0.4 S(w_i | w_{i-n+2}^{i-1}) & \text{otherwise} \end{cases}$$

S = Score

$$S(w_i) = \frac{C(w_i)}{|V|}$$

Other challenges

- Efficient storage
- Efficient lookup

<https://www.aclweb.org/anthology/D07-1090.pdf>

Beyond n-grams

Other types of language models

- Discriminative models:
 - train n-gram probabilities to directly maximize performance on end task (e.g. as feature weights)
- Parsing-based models
 - handle syntactic/grammatical dependencies
- Topic models
- Neural Language Models

Summary: Estimating language models

- ▶ Predict probability of sequence of words
- ▶ Need to handle **data sparsity**
use Markov assumption and smoothing
Independence assumptions
- ▶ Later: Neural language models

Reallocate probability mass

Ensure proper probability

Use Chain rule and approximate using a neural network

$$p(w_1, \dots, w_n) \approx \prod_t p(w_{t+1} \mid \underbrace{\phi(w_1, \dots, w_t)}_{\text{capture history with vector } s(t)})$$

How well do these models perform?

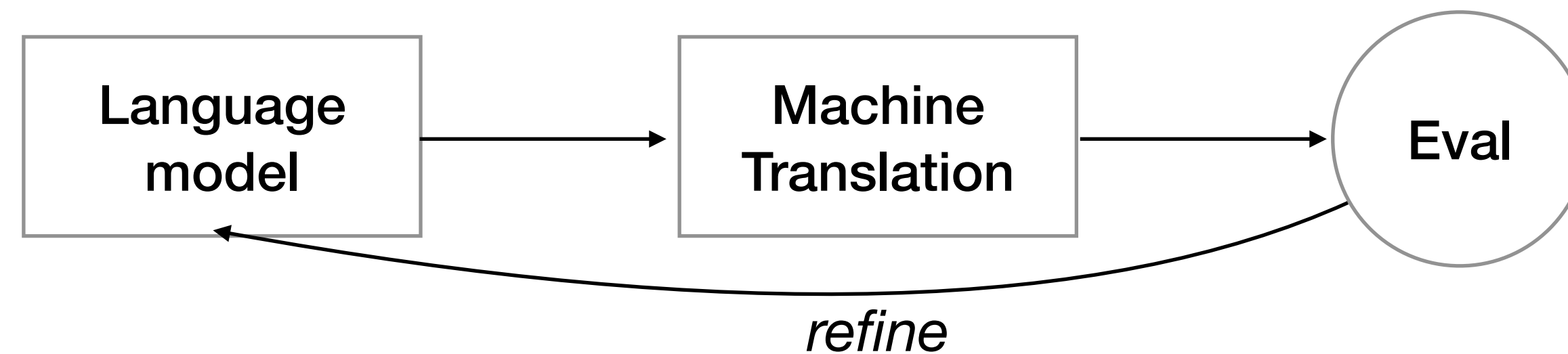
Evaluating Language Models

Evaluation

- **Extrinsic**: measure how useful the language model is at some task (MT, ASR, etc).
- **Intrinsic**: measure how good we are at modeling language

Extrinsic evaluation

- Train LM -> apply to task -> observe accuracy



- Directly optimized for downstream tasks
 - higher task accuracy -> better model
- Expensive, time consuming
- Hard to optimize downstream objective (indirect feedback)

Evaluating language models

- A good language model should assign higher probability to typical, grammatically correct sentences
- Research process:
 - **Train** parameters on a suitable training corpus
 - Assumption: observed sentences $\sim$ good sentences
 - **Test** on *different, unseen* corpus
 - Training on any part of test set not acceptable!
 - **Evaluation metric**



Evaluation of language models

Computing the **average** probability of the test corpus

- Given a test corpus $T = s_1, \dots, s_m$ of independent sentences, the probability of $P(T)$ is:

$$P(T) = \prod_{i=1}^m P(s_i) \quad \text{higher } P(T) = \text{better LM}$$

- But T can be any size and $P(T)$ will be lower if T is larger.
- So let's compute the **average** probability. Let M be the total number of tokens in the test corpus T .

$$M = \sum_{i=1}^m \text{length}(s_i)$$

- The average **log** probability of the test corpus T is:

$$\ell = \frac{1}{M} \log_2 \prod_{i=1}^m P(s_i) = \frac{1}{M} \sum_{i=1}^m \log_2 P(s_i)$$

Evaluation of language models

Perplexity

M = total # of words

m = # of sentences

- The average log probability of the test corpus T is:

$$\ell = \frac{1}{M} \sum_{i=1}^m \log_2 P(s_i)$$

higher ℓ = better LM

Note that ℓ is
a negative number

- Language models are evaluated using **perplexity**

$$\text{ppl}(T) = 2^{-\ell} \quad \text{lower ppl} = \text{better LM}$$

$\text{ppl}(T)$ is
a positive number

Note that the exponent ($-\ell$) can be regarded as the **cross entropy** between the empirical distribution of test corpus and the language model

Intuition: **Measure of model's uncertainty about next word**

Perplexity summary

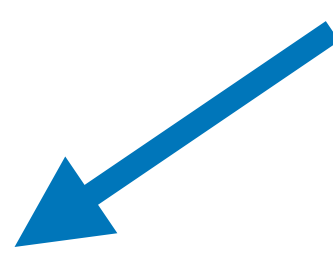
- Measure of how well a probability distribution (or model) predicts a sample

- For a corpus T with sentences $s_1, s_2, \dots, s_m$

$$\text{ppl}(T) = 2^{-\ell} \text{ where } -\ell = -\frac{1}{M} \sum_{i=1}^m \log_2 P(s_i)$$

where M is the total number of words in test corpus

cross entropy
between the
empirical
distribution of test
corpus and the
language model



- Unigram model: $-\ell = -\frac{1}{M} \sum_{i=1}^m \sum_{j=1}^{n_i} \log_2 P(w_j^{(i)}) = -\sum_{w \in V} \frac{C(w)}{M} \log_2 P(w)$
- Minimizing perplexity ~ maximizing probability

Intuition: Measure of model's uncertainty about next word
branching factor

Pros and cons of perplexity

Pros	Cons
Easy to compute	Domain match between train and test
standardized	Limited to sequence models
directly useful, easy to use to correct sentences	might not correspond to end task optimization
nice theoretical interpretation - matching distributions	$\log 0$ undefined
	can be cheated by predicting common tokens
	size of test set matters
	can be sensitive to low prob tokens/ sentences

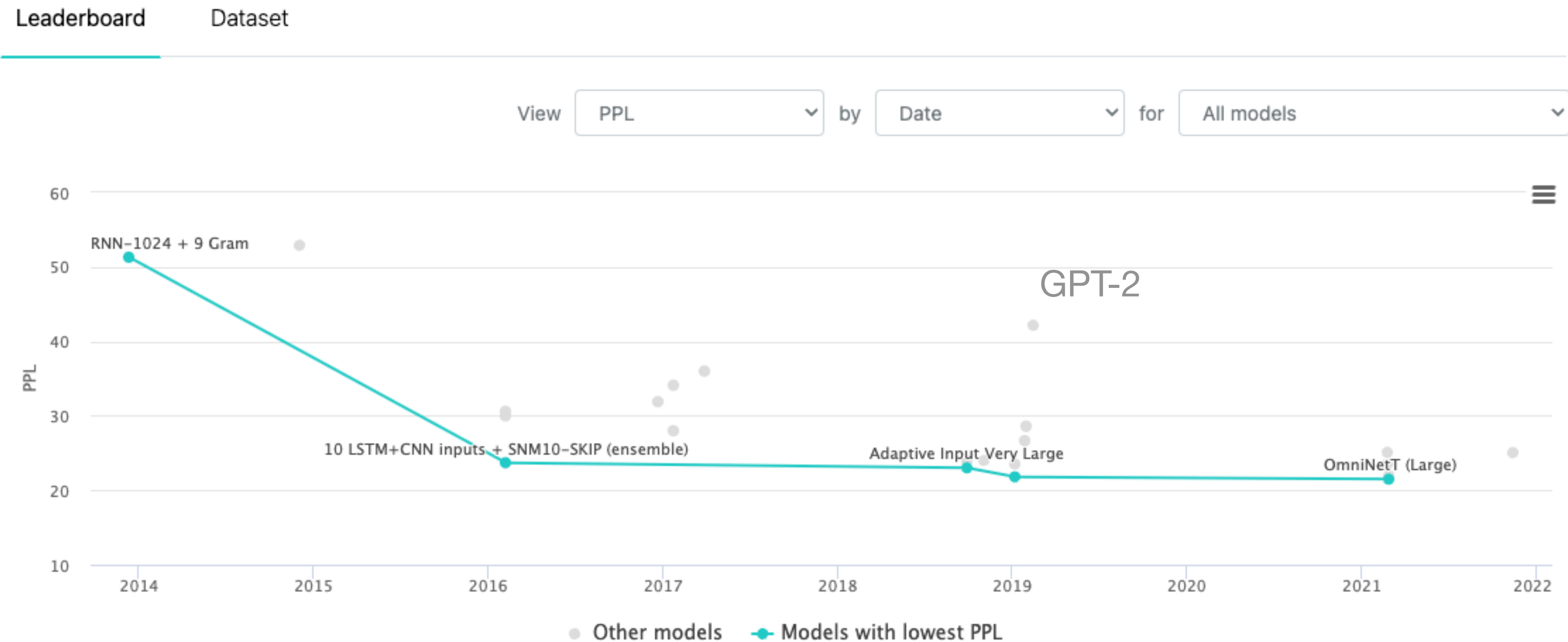
Perplexity values for different language models

Progress on the 1B Word Benchmark

Model	Params	Perplexity	Citation
unigram	775K	955	Chelba+ 2013
bigram	1B	137	Chelba+ 2013
trigram	1B	74	Chelba+ 2013
interpolated 5-gram	1.76B	67.6	Chelba+ 2013
10skip-gram+SNM	33B	52.9	Shazeer+ 2014
RNN-256 + 9-grams	20B	58.3	Chelba+ 2013
RNN-1024 + 9-grams	20B	51.3	Chelba+ 2013
Big LSTM+CNN	1.04B	30	Jozefowicz+ 2016
10 LSTMs+10skip-SNM	43B	23.7	Jozefowicz+ 2016
GPT2	1.54B	42.16	Radford+ 2019
Transformer XL	1.04B	21.8	Dai+ 2019
OmniNet	100M	21.5	Tay+ 2021

Perplexity of current LMs

Language Modelling on One Billion Word



<https://paperswithcode.com/sota/language-modelling-on-one-billion-word>

For other datasets: see <https://paperswithcode.com/task/language-modelling>

Where are we now?

Neural models with lots of data!

300 Billion tokens

Training data: mix of web + books + Wikipedia

Model Name	n_{params}
GPT-3 Small	125M
GPT-3 Medium	350M
GPT-3 Large	760M
GPT-3 XL	1.3B
GPT-3 2.7B	2.7B
GPT-3 6.7B	6.7B
GPT-3 13B	13.0B
GPT-3 175B or “GPT-3”	175.0B

Dataset	1B Word Benchmark test set	Quantity (tokens)	Weight in training mix	Epochs elapsed when training for 300B tokens
Common Crawl (filtered)		410 billion	60%	0.44
WebText2		19 billion	22%	2.9
Books1		12 billion	8%	1.9
Books2		55 billion	8%	0.43
Wikipedia		3 billion	3%	3.4

Open AI's GPT 3

Language Models are Few-Shot Learners
(Brown et al, 2020)

<https://arxiv.org/pdf/2005.14165.pdf>

Setting	PTB
SOTA (Zero-Shot)	35.8 ^a
GPT-3 Zero-Shot	20.5

Why the stop symbol is important?

Computing the probability of a sentence

Apply the Chain Rule: the trigram model

$$\begin{aligned} P(w_1, \dots, w_n) \\ &\approx P(w_1)P(w_2 \mid w_1)P(w_3 \mid w_1, w_2) \dots P(w_n \mid w_{n-2}, w_{n-1}) \\ &\approx P(w_1)P(w_2 \mid w_1) \prod_{i=3}^n P(w_i \mid w_{i-2}, w_{i-1}) \end{aligned}$$

- Notice that the length of the sentence n is variable
- What is size of the event space (e.g. the total number of possible events/sentences)?

Variable length sequences

Let $V = \{a, b\}$ and the language L be V^*

Consider a unigram model: $P(a) = P(b) = 0.5$

So strings in this language L are:

$a \quad 0.5$

$b \quad 0.5$

$aa \quad 0.5^2$

$bb \quad 0.5^2$

$\dots$

The sum over all strings in L should be equal to 1

But $P(a) + P(b) + P(aa) + P(bb) = 1.5!!!$

The **stop** symbol

What went wrong? We need to model variable length sentences

Add an explicit probability for the stop symbol

$$P(a) = P(b) = 0.25 \quad P(\text{stop}) = 0.5$$

Now strings have the following probabilities:

$$\begin{array}{ll} \text{stop} & 0.5 \\ a \text{ stop} & 0.25 \times 0.5 = 0.125 \\ b \text{ stop} & 0.25 \times 0.5 = 0.125 \\ aa \text{ stop} & 0.25^2 \times 0.5 = 0.03125 \\ bb \text{ stop} & 0.25^2 \times 0.5 = 0.03125 \\ & \dots \end{array}$$

The sum is no longer greater than one!

The stop symbol

- With this new stop symbol, we can show that $\sum_{u \in L} P(u) = 1$
- Let $p_s = P(\text{stop})$, the probability of the stop symbol
- Then, we can show that the probability of all sequences of length n is $p(n) = p_s(1 - p_s)^n$

$$\begin{aligned}
 p(n) &= \sum_{w_1, \dots, w_n} p(w_1, \dots, w_n) \times p_s \text{ where } w_j \neq \text{stop} \\
 &= p_s \sum_{w_1} \dots \sum_{w_n} p(w_1, \dots, w_n) \\
 &= p_s \sum_{w_1} \dots \sum_{w_n} p(w_1) \dots p(w_n) \\
 &= p_s \sum_{w_1} p(w_1) \dots \sum_{w_n} p(w_n) \\
 &= p_s \prod_{j=1}^n \sum_{w_j} p(w_j) \\
 &= p_s (1 - p_s)^n
 \end{aligned}$$

$\sum_{w \neq \text{stop}} P(w) = 1 - p_s$


The stop symbol

- With this new stop symbol, we can show that $\sum_{u \in L} P(u) = 1$
- Let $p_s = P(\text{stop})$, the probability of the stop symbol
- Using that the probability of all sequences of length n is $p(n) = p_s(1 - p_s)^n$

$$\begin{aligned}\sum_{u \in L} P(u) &= \sum_{n=0}^{\infty} p(n) = \sum_{n=0}^{\infty} p_s(1 - p_s)^n \\ &= p_s \sum_{n=0}^{\infty} (1 - p_s)^n \\ &= p_s \frac{1}{1 - (1 - p_s)} = p_s \frac{1}{p_s} = 1\end{aligned}$$

Summary

- Language models estimates the probability of a sentence
- Statistical LMs: N-grams: $P(w_i|w_{i-1}) = \frac{C(w_{i-1}, w_i)}{C(w_{i-1})}$
- Smoothing to handle data sparsity
- Perplexity for evaluating language models
- Modern NLP powered by neural-based language models

Reminders

- HW-0 due next Wednesday 11/17
 - Submit via gradescope and coursys
- Next week:
 - Classification for NLP